SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: COMPILER DESIGN**

COURSE OUTCOME COVERED

CO5: *Develop code optimization and Code Generation using DAG representation with data flow analysis to produce optimized target code.* *(BL3)*

Assessment Tool Weight-age: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

SIMULATION TASK

Code of Conduct:

I Ramya S S/192421226 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ramya S S

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts

Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
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Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

SIMULATION TASK

SIMULATION TASK – COMPILER DESIGN

Aim:

To simulate compiler optimization techniques, basic block formation, DAG construction, register allocation, and target code generation.

1. Peephole Optimization

Given Code
t1 = x * 1
t2 = t1 + y
t3 = t2 - 0
t4 = t3 * 0
if t4 != 0 goto L1
t5 = y + y

Step 1: Algebraic Simplification
Using algebraic identities:
x * 1 = x
t2 - 0 = t2
Therefore:
t1 = x
t2 = t1 + y
t3 = t2
t4 = t3 * 0
if t4 != 0 goto L1
t5 = y + y

Step 2: Constant Folding
Since:
t3 * 0 = 0
we get:
t4 = 0
if 0 != 0 goto L1
The condition is always false, so the jump is removed.

Step 3: Dead Code Elimination
t1, t2, t3, and t4 do not contribute to the final useful computation. Hence they are removed.
Optimized Code
t5 = y + y
Optimizations Used

Optimization	Example
Algebraic simplification	x*1 → x, x-0 → x

Constant folding	**t3*0 → 0**
Dead code elimination	**Removes unused temporaries**
Constant condition	**if 0!=0 removed**

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2. Basic Blocks and Control Flow Graph

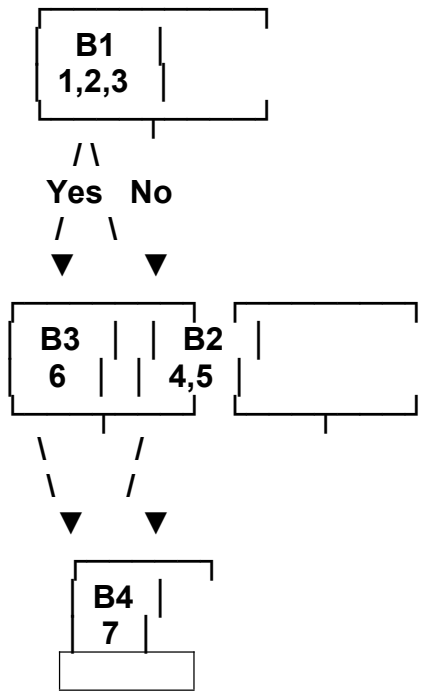
Given Code
(1) x = a + b
(2) y = x * c
(3) if y > 50 goto 6
(4) z = y - d
(5) goto 7
(6) z = y + d
(7) w = z * 2

Rules for Leaders
A statement is a leader when:

1. **It is the first statement.**
2. **It is the target of a jump.**
3. **It immediately follows a jump statement.**

Leaders
1, 4, 6, 7
Basic Blocks
B1
1. x = a + b
2. y = x * c
3. if y > 50 goto 6
B2
4. z = y - d
5. goto 7
B3
6. z = y + d
B4
7. w = z * 2

Control Flow Graph



CFG Edges
B1 → B3 (y > 50)
B1 → B2 (y ≤ 50)
B2 → B4
B3 → B4

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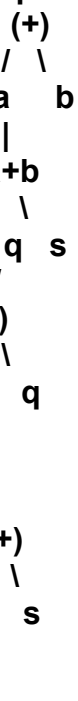
3. DAG and Common Sub-Expression

Given Basic Block
p = a + b
q = a + b
r = p * q
s = a + b
t = r + s

Common Sub-Expression
The expression:
a + b
occurs three times:
p = a + b
q = a + b
s = a + b
Therefore, it should be calculated only once.

DAG Representation



Optimized Intermediate Code

```
p = a + b
q = p
r = p * q
s = p
t = r + s
```

Target Code Using Two Registers

```
LOAD R0, a
ADD  R0, b
MOV  R1, R0
MUL  R0, R1
ADD  R0, R1
STORE t, R0
```

Register Allocation

Step	R0	R1
Load a	a	—
Compute a+b	a+b	—
Save common value	a+b	a+b
Multiply	$(a+b)^2$	a+b
Final addition	$(a+b)^2+(a+b)$	a+b

Thus, the repeated a+b calculation is eliminated.

4. Code Generation with Two Registers

Given Code
t1 = a + b
t2 = c * d
t3 = t1 - t2
t4 = e + f
t5 = t3 / t4

Registers available:
R0, R1
Target Code
LOAD R0, a
ADD R0, b

LOAD R1, c
MUL R1, d

SUB R0, R1

LOAD R1, e
ADD R1, f

DIV R0, R1
STORE t5, R0

Register Descriptor and Address Descriptor

Step	R0	R1	Address Descriptor
Initial	Empty	Empty	Variables in memory
t1=a+b	t1	Empty	t1 → R0
t2=c*d	t1	t2	t1 → R0, t2 → R1
t3=t1-t2	t3	t2	t3 → R0, t2 → R1
t4=e+f	t3	t4	t3 → R0, t4 → R1
t5=t3/t4	t5	t4	t5 → R0, t4 → R1

Explanation

- **R0 stores the result of a+b.**
- **R1 stores c*d.**
- **After subtraction, R0 contains t3.**
- **R1 is reused for e+f.**
- **Finally, R0/R1 produces t5.**

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5. RISC Target Code Generation
Expression
(a - b) * (c + d) - e
Target Instructions
First calculate (a-b):
LOAD R0, a
LOAD R1, b

SUB R0, R1
Now calculate (c+d):
LOAD R2, c
LOAD R3, d
ADD R2, R3
Multiply the two results:
MUL R0, R2
Subtract e:
LOAD R1, e
SUB R0, R1
Store the final result:
STORE result, R0
Complete Code
LOAD R0, a
LOAD R1, b
SUB R0, R1

LOAD R2, c
LOAD R3, d
ADD R2, R3

MUL R0, R2

LOAD R1, e
SUB R0, R1

STORE result, R0

Additional Given Code – Peephole Optimization

Given
t1 = a + b
t2 = t1 + 0
t3 = t2 * 1
if t3 == 0 goto L1
t4 = t3 + t3
Using:
x + 0 = x
x * 1 = x
we get:
t1 = a + b
t2 = t1
t3 = t2
if t3 == 0 goto L1
t4 = t3 + t3

RUBRICS FOR CALCULATION

Simulation tools

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